What is chemistry?

Matter and Energy

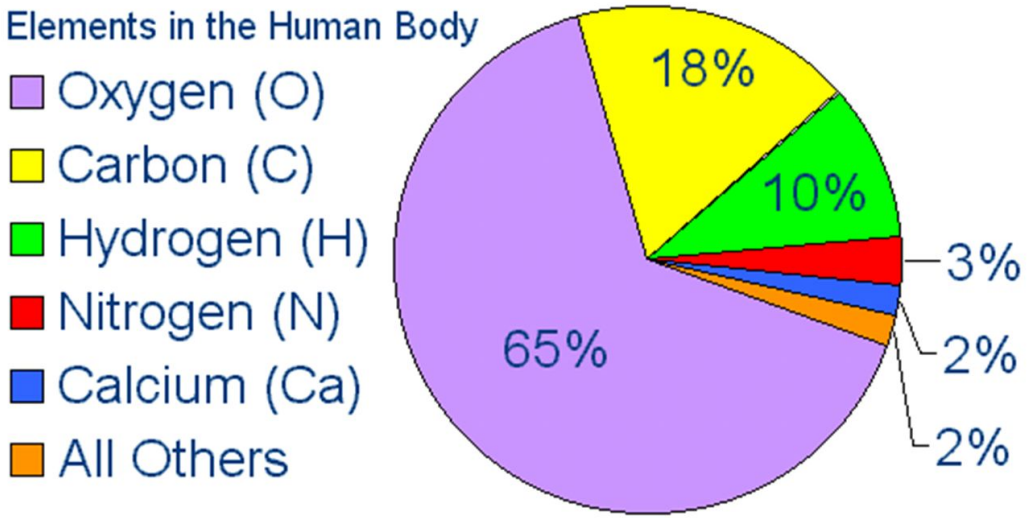
- Everything in the universe can be classified as either **energy** or **matter**
- **Energy** is the ability to perform **work** or make things **move** (e.g. heat, sound, light etc.)
- **Matter** is anything that has **mass** and **volume** (takes up space)

Chemistry & Chemicals

- Chemistry is the study of **matter**

What is matter made of?

- **Chemicals**



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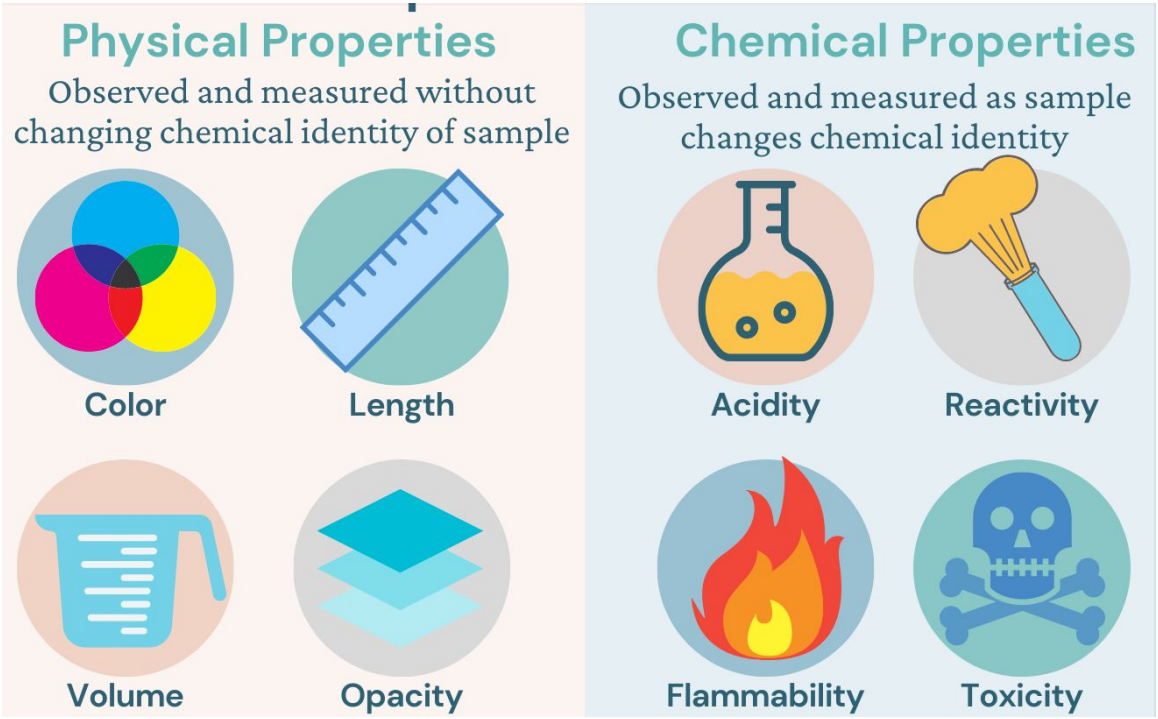
Describing Matter

- One of the first things chemists must be able to do is to **describe** matter
- This can be done by observing its **physical** and **chemical** properties



Physical Properties

- A **physical** property of matter is a characteristic that can be observed **without** producing a **new** substance
- The bar is **hard**, **lustrous**, **smooth**, and is **yellow** in colour
- It has a **mass** of **11.34 kg**

Chemical Properties

- Chemists can also observe how a substance **behaves** when **mixed** with other substances
- That is, how it **reacts** to produce **new** substances
- When an apple browns, it is **reacting** with the oxygen in the air

Your turn

- Properties of Matter Vocabulary
- Physical vs. Chemical Properties Practice